

Assignment 2

Submission deadline: 15th June 2026, 21:59

General instructions

- Please type your answers in Word or L^AT_EX and submit your answer sheet in **PDF format**. Handwriting will not be graded.
- For programming questions, submit your Python code in **.py files**, not Jupyter notebooks.
- Please do **not** zip your files when uploading to BrightSpace. Zipped packages will not be graded.
- Group discussion is **not** allowed.
- The grade is subject to review during the Oral exam.
- **Late submission policy:** A penalty of **1 point per 30 minutes** (or part thereof) will be deducted for late submissions.
- A final grade below **50%** (i.e. below 10 out of 20 points) can use the resit to repair the grade to a **maximum of 60%** of the full points (i.e. at most 12 out of 20 points).

General notation

The following notation is used throughout this assignment:

$S_i, S_i(t)$	underlying stock (asset) price of asset i ; $S_i(t)$ denotes its price at time t .
$x_i(t)$	log-price $x_i(t) = \ln S_i(t)$ (used in characteristic-function based methods).
w_i	weight of asset i in the basket; $\sum_i w_i = 1, w_i \geq 0$.
B_T	basket value at maturity, $B_T = \sum_{i=1}^n w_i S_i(T)$.
K	strike price.
T	maturity (in years).
r	continuously compounded risk-free interest rate.

ρ instantaneous correlation between the two driving Brownian motions W_1, W_2 .

$\varphi(\cdot)$ characteristic function (ChF) of the relevant state variable.

Common contract specification. In all three exercises we price the European basket call

$$V(0) = e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[\max(w_1 S_1(T) + w_2 S_2(T) - K, 0) \right],$$

with $n = 2$ underlyings, weights $w_1 = w_2 = \frac{1}{2}$, maturity $T = 1$ year, risk-free rate $r = 0.02$, no dividends, and strike $K = 100$ (at-the-money).

Note that option should be priced under the risk neutral measure, where the discounted asset price is a martingale. So do remember the change of the measure from the physical measure to the risk neutral measure when you derive the dynamics of the underlying assets, and make sure to use the risk neutral dynamics in your implementation.

Technical Hints

For a square-root diffusion process of the form

$$dS_t = rS_t dt + \sigma \sqrt{S_t} dW_t,$$

where $S_0 > 0$ and W_t is a standard Brownian motion, the terminal distribution of S_T follows

$$S_T \sim e^{rT} \cdot \frac{\sigma^2 \cdot \tau(T)}{4} \cdot Z,$$

with $\tau(T) = \frac{1-e^{-rT}}{r}$ and Z is a non-central chi-squared distribution with zero-degrees of freedom and non-centrality parameter $\lambda = \frac{4S_0}{\sigma^2 \tau(T)}$, i.e.,

$$Z \sim \chi_0'^2(\lambda).$$

1. Basket Options via Monte Carlo Simulation (6 points)

The risk-neutral price of the basket call is the discounted expectation

$$V(0) = e^{-rT} \mathbb{E}^{\mathbb{Q}} \left[\max(w_1 S_1(T) + w_2 S_2(T) - K, 0) \right]. \quad (1)$$

In this question you are asked to implement the Monte Carlo (MC) estimator of the basket option price under two different assumptions on the underlying dynamics, and to design convergence tests for both cases. The two cases are as follows.

Note that exact simulation is required if available. If exact simulation is not available, you may use a time-stepping scheme of your choice (e.g. Euler scheme etc.) and design a separate convergence test for both the time-discretisation error and the simulation error with respect to the number of simulation paths.

(a) Both underlyings log-normal.

Assume both assets follow geometric Brownian motion under the physical measure:

$$\begin{aligned} dS_1(t) &= \mu S_1(t) dt + \sigma_1 S_1(t) dW_1(t), \\ dS_2(t) &= \mu S_2(t) dt + \sigma_2 S_2(t) dW_2(t), \quad dW_1 dW_2 = \rho dt. \end{aligned}$$

Parameter values in the physical measure $S_1(0) = 100$, $S_2(0) = 100$, $\mu = 0.1$, $\sigma_1 = 0.20$, $\sigma_2 = 0.30$, $\rho = 0.30$.

Document your Monte carlo simulation scheme. After that, implement the MC estimator and compute a 95% confidence interval (CI) for the price (2 points).

Design convergence tests to check if the implementation is correct. Show the convergence graph and the convergence table, including the calculated convergence rate (1 point).

(b) One log-normal, one CIR.

Now assume that under the physical measure the second underlying follows a Cox–Ingersoll–Ross (square-root) process while the first remains log-normal:

$$\begin{aligned} dS_1(t) &= \mu S_1(t) dt + \sigma_1 S_1(t) dW_1(t), \\ dS_2(t) &= \kappa (\theta - S_2(t)) dt + \sigma_2 \sqrt{S_2(t)} dW_2(t), \quad dW_1 dW_2 = \rho dt. \end{aligned}$$

The CIR specification keeps $S_2(t) \geq 0$ provided the Feller condition $2\kappa\theta \geq \sigma_2^2$ holds.

Parameter values in the physical measure $S_1(0) = 100$, $\sigma_1 = 0.25$, $S_2(0) = 100$, $\mu = 0.1$, $\kappa = 0.015$, $\theta = 100$, $\sigma_2 = 1.5$, and $\rho = 0.1$.

Document your Monte carlo simulation scheme. After that, implement the MC estimator and compute a 95% confidence interval (CI) for the price (2 points).

Design convergence tests to check if the implementation is correct. Show the convergence graph and the convergence table, including the calculated convergence rate (1 point).

2. Basket Options via Moment Matching (6 points)

An analytical approximation is the *moment-matching* method: approximate the basket distribution by a distribution whose first few moments match those of B_T (the basket value at maturity), and price the basket option under the approximating distribution.

Let $m_k = \mathbb{E}[B_T^k]$ denote the k -th moment of B_T .

- (a) **Both underlyings log-normal. See question 1(a).**

Parameter values in the physical measure $S_1(0) = 100$, $S_2(0) = 100$, $\mu = 0.1$, $\sigma_1 = 0.20$, $\sigma_2 = 0.30$, $\rho = 0.30$.

Choose a distribution to match the target moments and price the basket option under the approximating distribution - document your approximation approach clearly with the technical details for the moments and approximating distribution, and implement it in Python (2 points). Design your error convergence analysis to show that matching higher order moments improves the accuracy of the approximation (1 point).

- (b) **One log-normal, one CIR. See question 1(b).**

Parameter values in the physical measure $S_1(0) = 100$, $\sigma_1 = 0.25$, $S_2(0) = 100$, $\mu = 0.1$, $\kappa = 0.015$, $\theta = 100$, $\sigma_2 = 1.5$, and $\rho = 0$.

Choose a distribution to match the target moments and price the basket option under the approximating distribution - document your approximation approach clearly with the technical details for the moments and approximating distribution, and implement it in Python (2 points). Design your error convergence analysis to show that matching higher order moments improves the accuracy of the approximation (1 point).

3. Basket Options via COS Method (8 points)

In this question you are asked to implement the COS method for the same two setups as in Questions 1 and 2, and to design convergence tests for both cases. Note that you can choose to use the one-dimensional COS method or the two-dimensional implementation, depending on how you formulate the problem. Read <https://ir.cwi.nl/pub/20771/SIAM.pdf> for the original paper on the two-dimensional COS method. And use numerical methods if you have to.

- (a) **Both underlyings log-normal. See question 1(a).**

Parameter values in the physical measure $S_1(0) = 100$, $S_2(0) = 100$, $\mu = 0.1$, $\sigma_1 = 0.20$, $\sigma_2 = 0.30$, $\rho = 0$.

Document your implementation clearly with the technical details for the derivation of the ch.f., COS coefficients, and the formula of the basket option under the COS method, and implement the COS method in Python (2 points). Justify your choice of the truncation domain and the number of Fourier coefficients (1 point). Design your error convergence analysis to show that the COS method converges with the expected convergence rate (1 point).

- (b) **One log-normal, one CIR. See question 1(b).**

Parameter values in the physical measure $S_1(0) = 100$, $\sigma_1 = 0.25$, $S_2(0) = 100$, $\mu = 0.1$, $\kappa = 0.015$, $\theta = 100$, $\sigma_2 = 1.5$, and $\rho = 0$.

Document your implementation clearly with the technical details for the derivation of the ch.f., COS coefficients, and the formula of the basket option under the COS method, and implement the COS method in Python (2 points). Justify your choice of the truncation domain and the number of Fourier coefficients (1 point). Design your error convergence analysis to show that the COS method converges with the expected convergence rate (1 point).